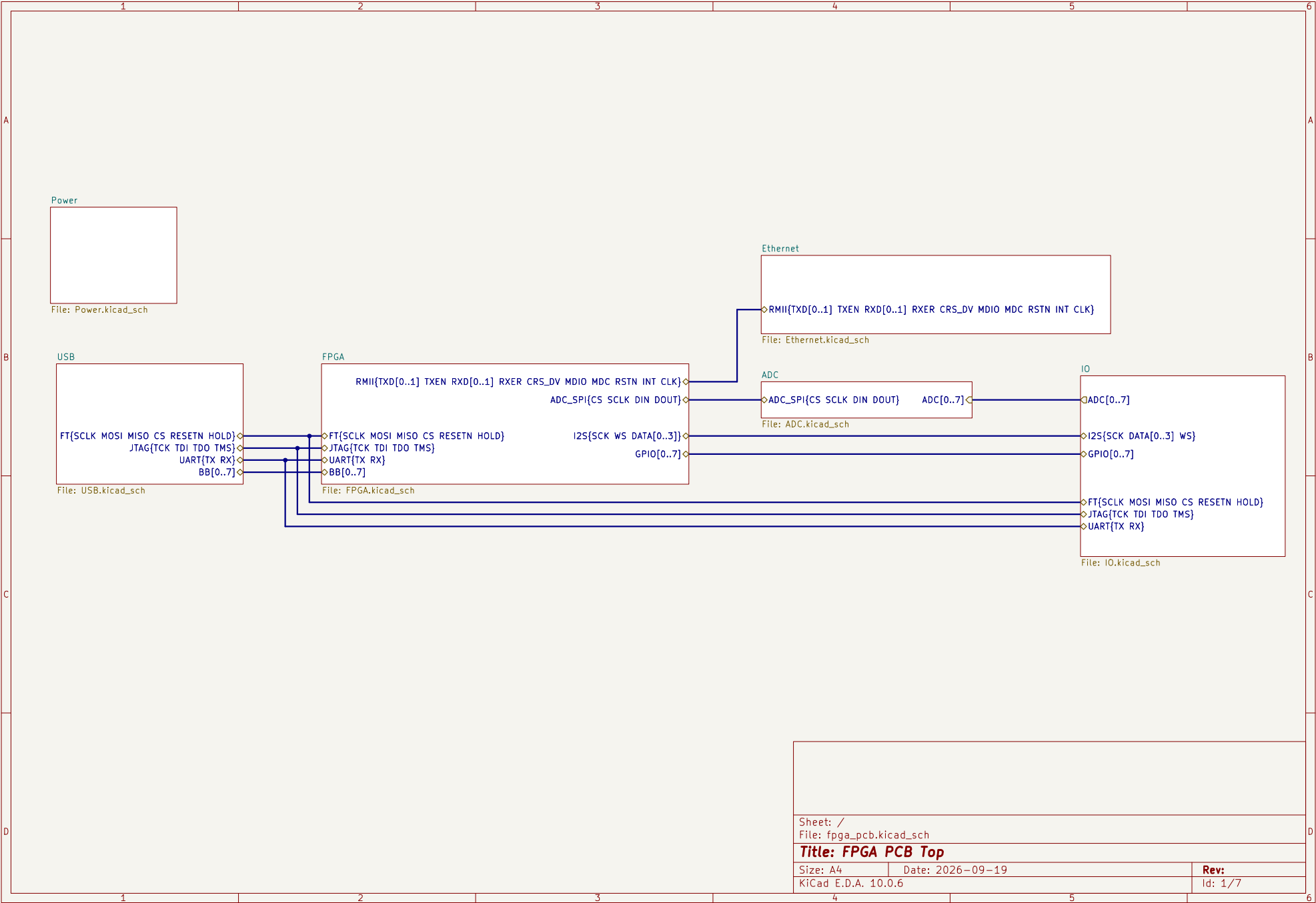
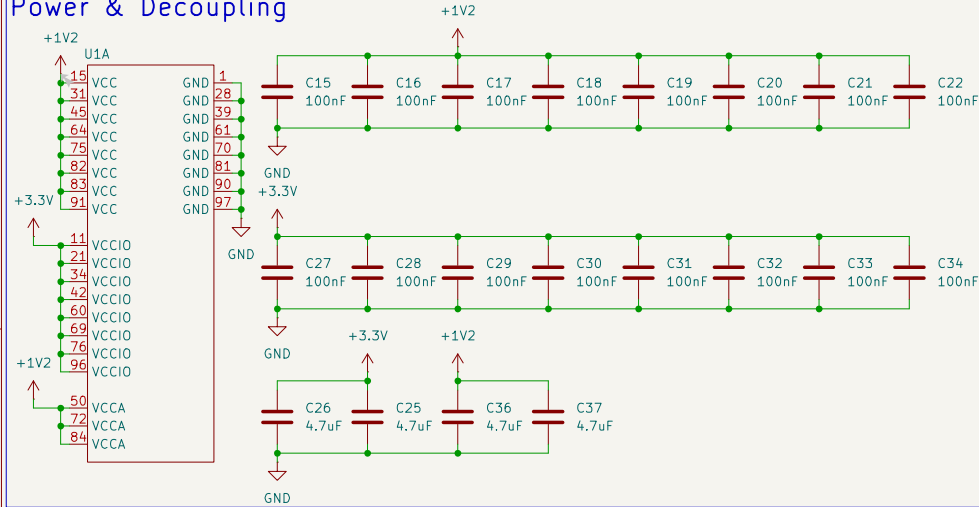
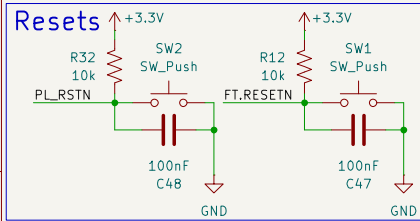




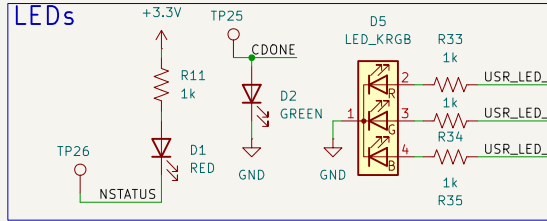
Power & Decoupling



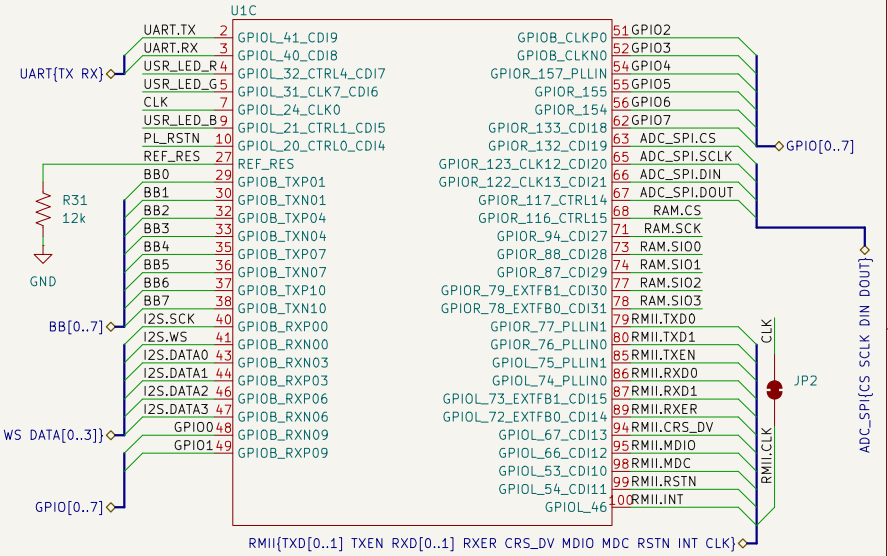
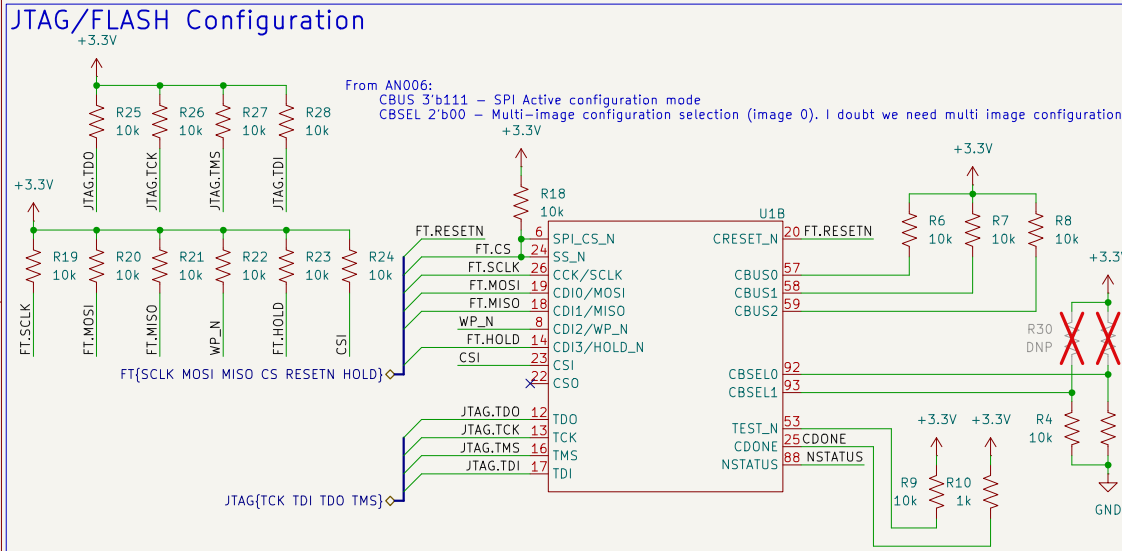
Resets



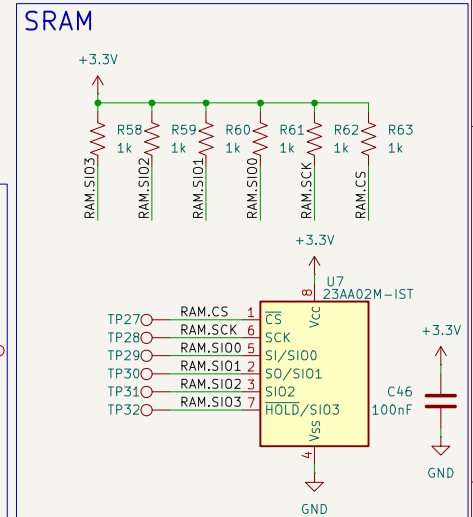
LEDs



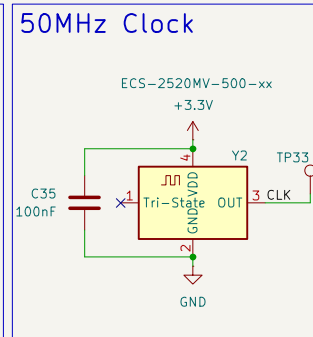
JTAG/FLASH Configuration



SRAM



50MHz Clock



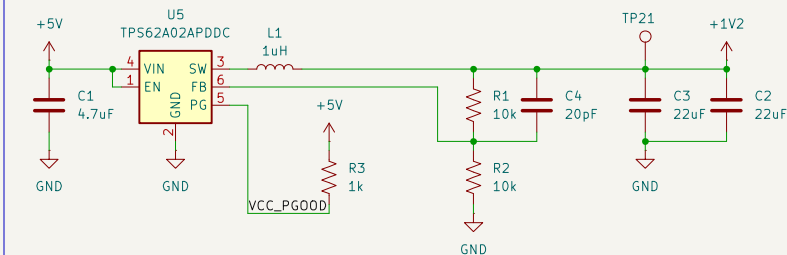
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File: FPGA.kicad_sch
Title: FPGA

Size: A4 Date: 2026-09-19
KiCad E.D.A. 10.0.6

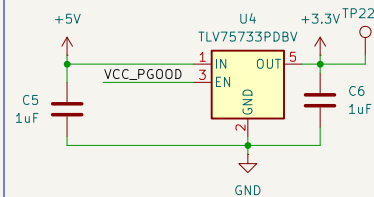
Rev: 2/7

Power estimator gives ~1.7W at 75% utilization, 300MHz, and 50% activity.
 Core power supply will be designed for 2A@1.2V.
 The T20 datasheet specifies +/- 50mV tolerance on VCC so ripple is a priority here.
 IO power supply can probably get away with a LDO - TODO check this
 T20 datasheet recommends bringing VCCIO up AFTER VCC.
 VCCIO rail is brought up with an enable pin from the PGOOD signal of VCC

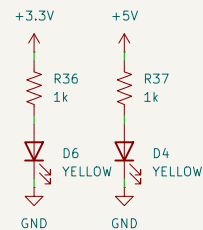
1.2V Core Regulator



3.3V IO Regulator

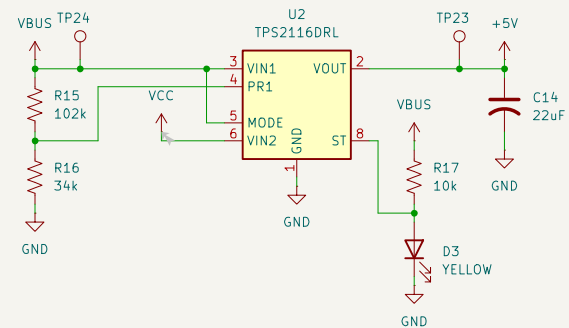


PWR LEDs



USB_5V/EXT_5V Switch

Power supply switch, prioritizes USB 5V.
 When USB drops below 4V, switches to VIN.
 PR1 reference voltage is 1V.
 ST is pulled low when VBUS is not in use.
 Indicator LED will turn on when USB 5V is available.
 From datasheet - switchover time is 8us.
 C out should smooth that out.
 Assuming current draw of 1.2A and min safe voltage of 4.5V
 $(5V - 4.5V) * C / 1.2A = 8us$
 $C = 12.2uF$ Round up to 22uF



Sheet: /Power/
 File: Power.kicad_sch

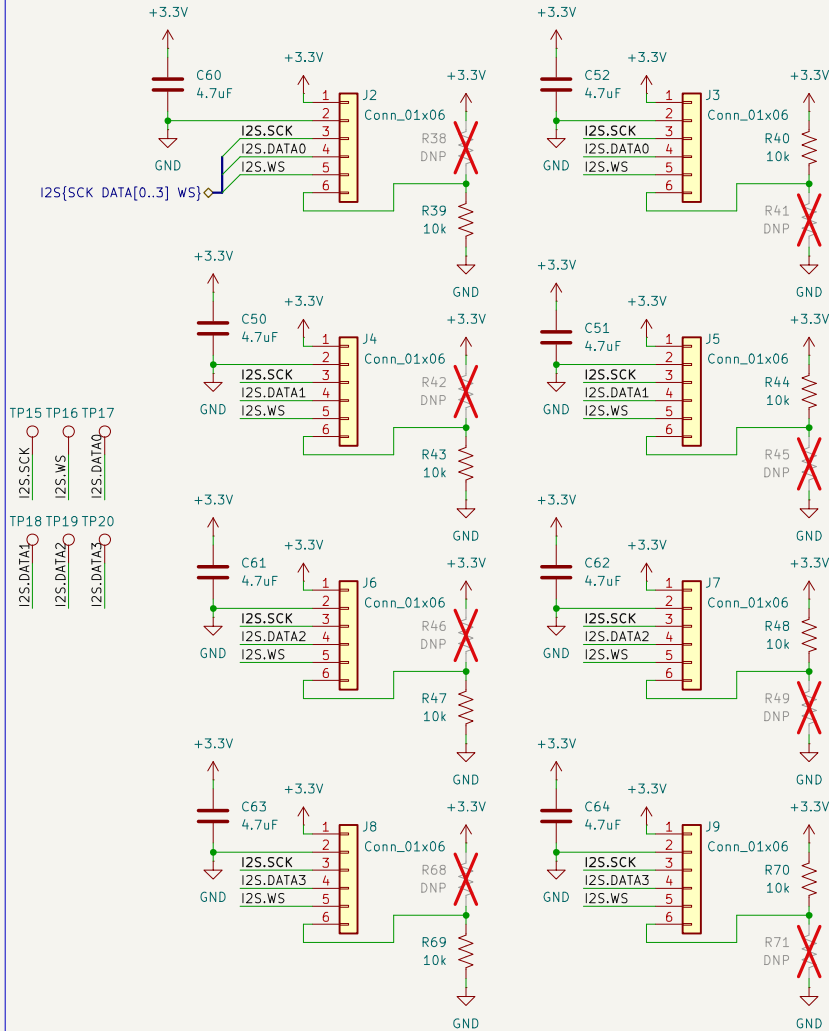
Title: Power

Size: A4 Date: 2026-09-19
 KiCad E.D.A. 10.0.6

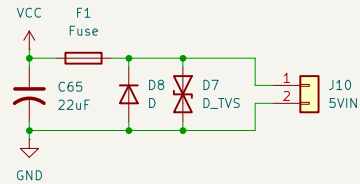
Rev:
 Id: 3/7

I2S Connectors

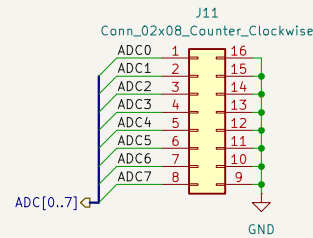
I2S shares clk and ws across all 8 microphones to save pins.
I2S also supports 2 channels, so each data line is shared across 2 mics, multiplexed by the WS signal.
Resistors are used as the SEL pin, which selects which channel a mic is on.



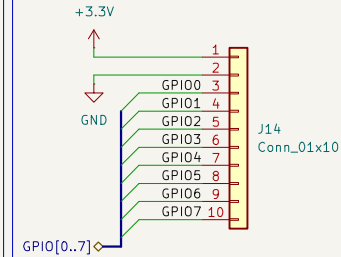
5V Input



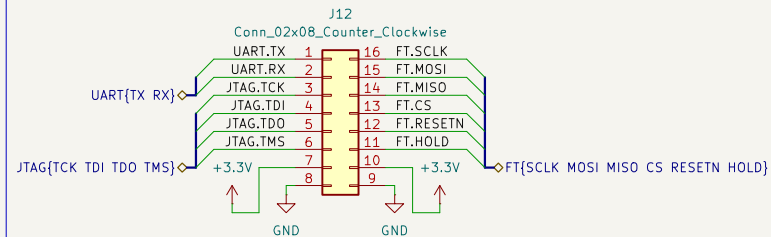
ADC Inputs



GPIO



Programming



Sheet: /IO/
File: IO.kicad_sch

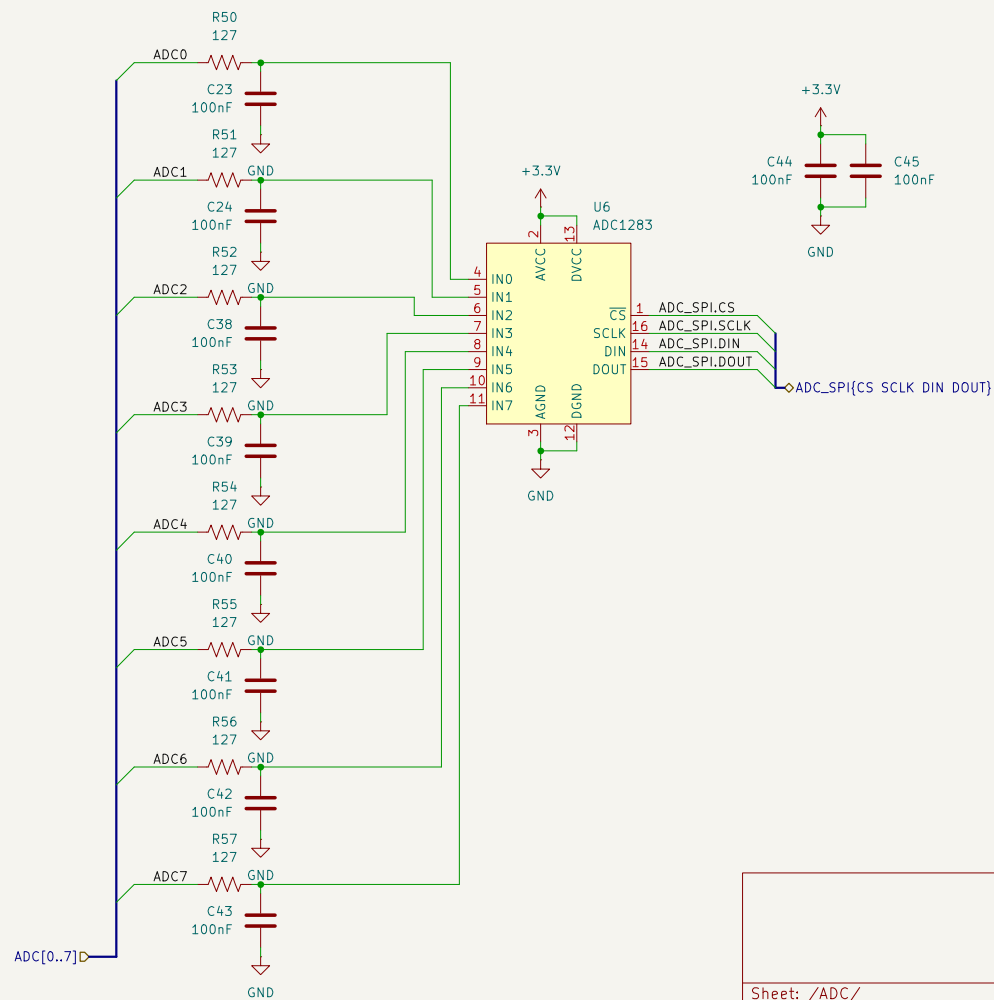
Title: I/O

Size: A4 Date: 2026-09-19

KiCad E.D.A. 10.0.6

Rev:
Id: 5/7

ADC has 8 channels with a total max fs of 200kHz.
Antialiasing filters have a cutoff frequency of ~12.5kHz
(200kHz/8/2)



Sheet: /ADC/
File: ADC.kicad_sch

Title: ADC

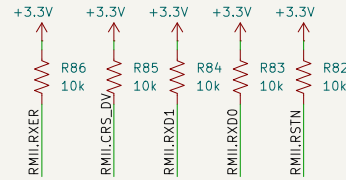
Size: A4 Date: 2026-09-19
KiCad E.D.A. 10.0.6

Rev:
Id: 6/7

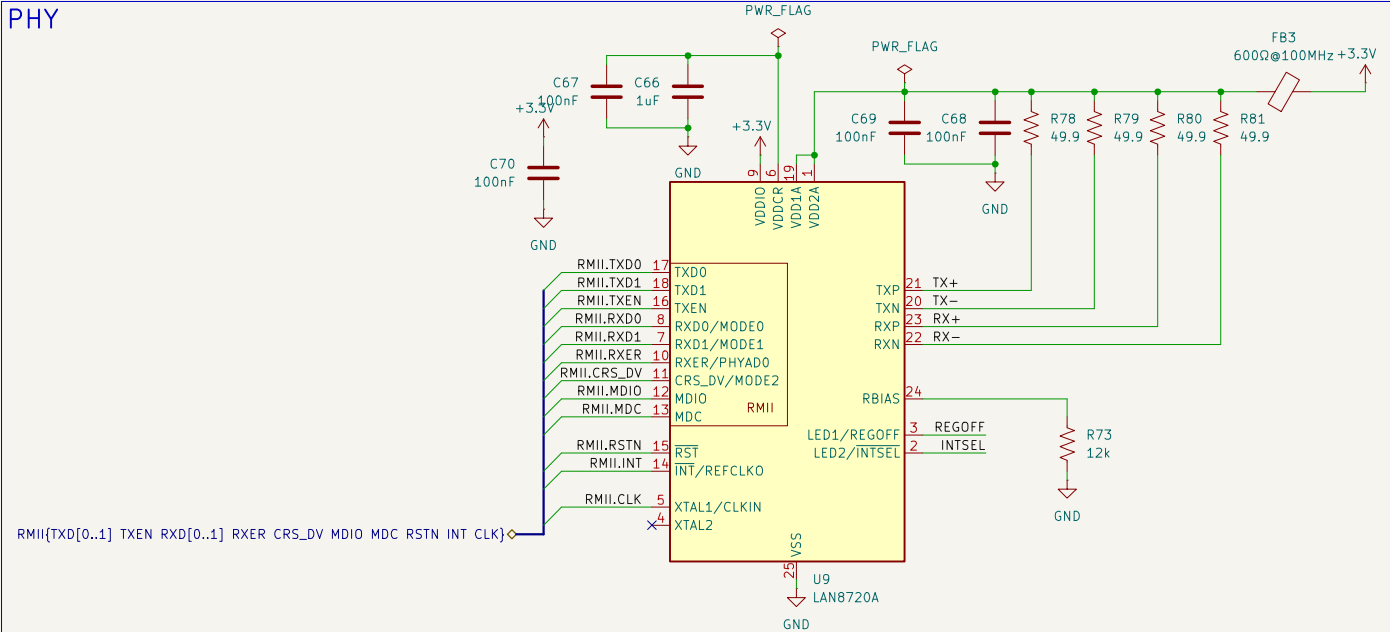
RMII Testpoints



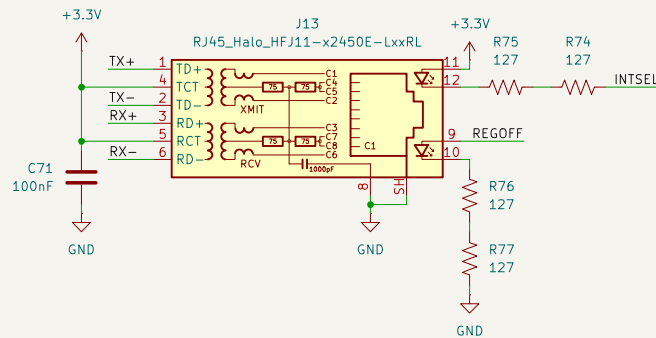
Config Pullups



PHY



RJ45



Sheet: /Ethernet/
File: Ethernet.kicad_sch

Title: Ethernet

Size: A4 Date: 2026-09-19
KiCad E.D.A. 10.0.6

Rev:
Id: 7/7